

Teaching critical thinking

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Preface

This is a Quarto book.

To learn more about Quarto books visit <https://quarto.org/docs/books>.

Part I

Critical thinking

1 Introduction

This is a book created from markdown and executable code.

See Knuth (1984) for additional discussion of literate programming.

2 How science works

3 What is theory?

4 Working with evidence

5 Lying, cheating, state capture, and life and death questions

6 Criticality in writing

7 Criticality in visualization

8 Criticality citation practice

9 Subjectivity, objectivity, and scientific virtues

10 Summary

In summary, this book has no content whatsoever.

Part II

Quarto

11 Setting up a book in Quarto

11.1 Introduction

We want the production of the book to be self-documenting. This will allow us to share Quarto process information. Opening chapters begin with book level stuff. Later chapters focus on within-chapter content writing and editing.

This chapter gets you from nothing to the skeleton structure of a book, published as live webpages through GitHub Pages.

11.2 New project, GitHub first

We are going to take an approach set out here (there are others). It helps us to understand how to work where there is an existing GitHub repository.

<https://happygitwithr.com/new-github-first>

11.2.1 Make a repo on GitHub

Go to <https://github.com> and make sure you are logged in.

You will see your profile page.

Near ‘Repositories’, click the big green ‘New’ button. Or, if you are on your own profile page, click on ‘Repositories’, then click the big green ‘New’ button.

11.2.2 Name the repository, add a description, keep it public, add a README, add .gitignore, add a license

You will have to enter some information:

- Repository template: No template.
- Repository name: myrepo or whatever you wish to name your new project. Approach this similar to a variable name, in code: descriptive but brief, no whitespace. Letters, digits, -, ., or _ are allowed.

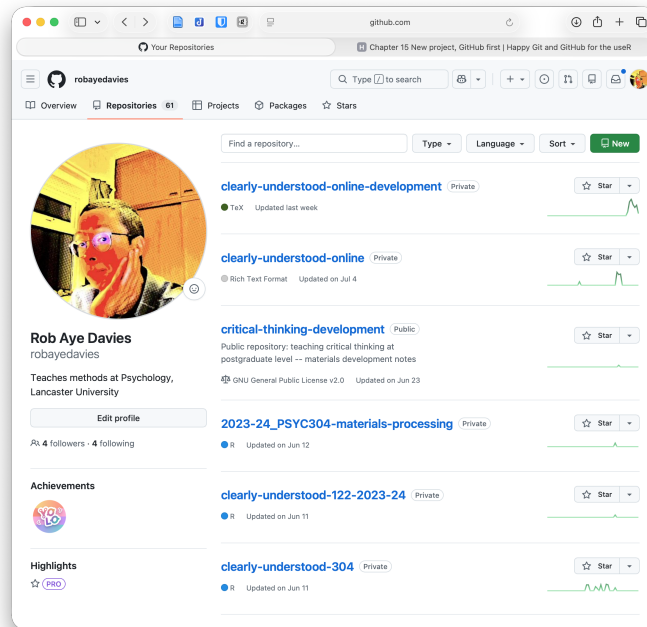


Figure 11.1: GitHub new repo

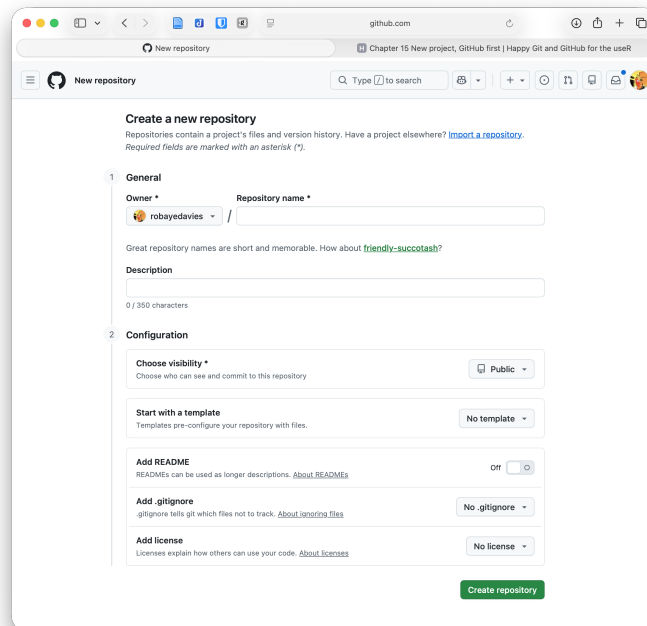


Figure 11.2: github-new-repo-name-public

- Description: any short description of the project.
- Public.
- Initialize this repository with: **Add a README file**.
- Click the big green button that says **Create repository**.

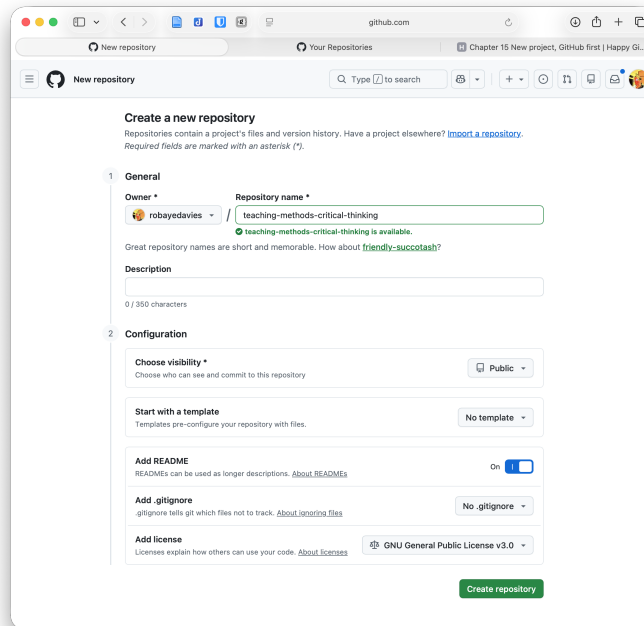


Figure 11.3: github-new-repo-name-public-values

Now click the big green button that says **<> Code**. Copy a clone URL to your clipboard. You will need this for the R-Studio side of the set-up.

11.2.3 Now, in RStudio, set up a project (GitHub repo first)

We are going to create a project, using version control.

File > New Project > Version control

We will be cloning a project from a Git repository: the repo we just created.

File > New Project > Version control > Git

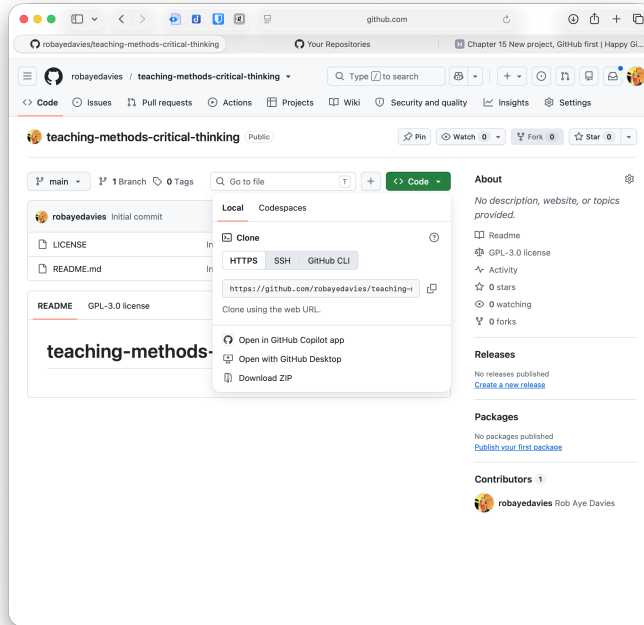
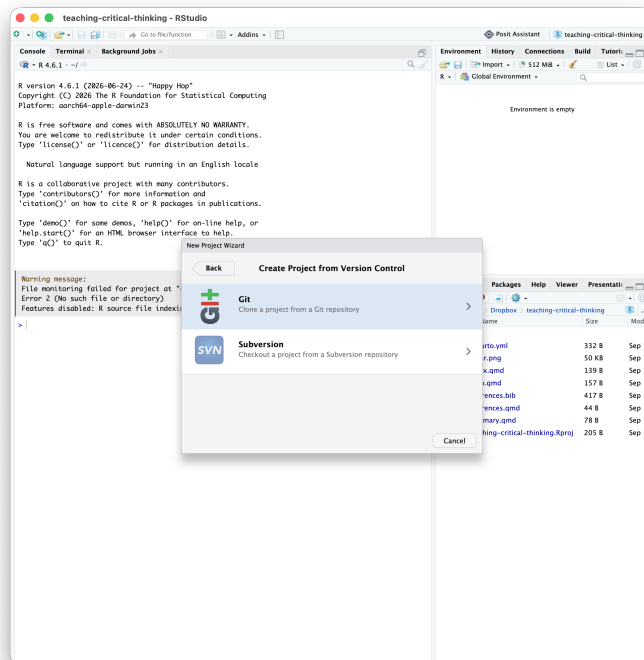


Figure 11.4: github-new-repo-url



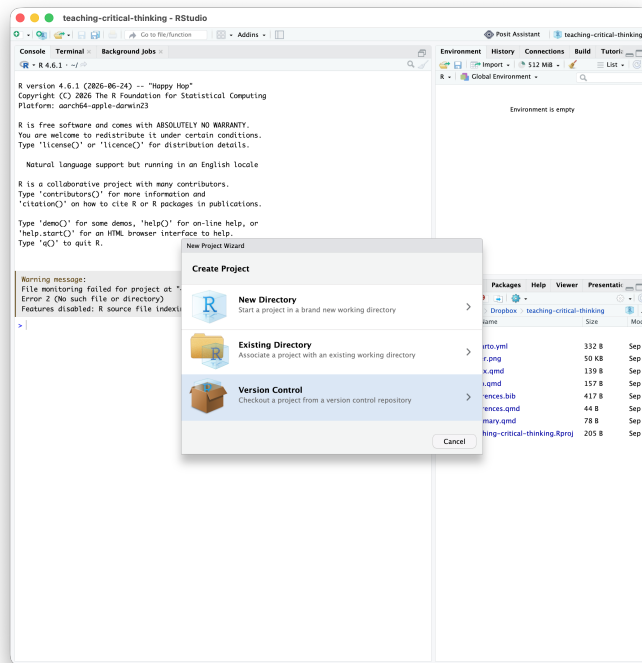


Figure 11.5: rstudio-create-project-version-control

Enter the URL from the GitHub repository

- Remember where we got the URL:

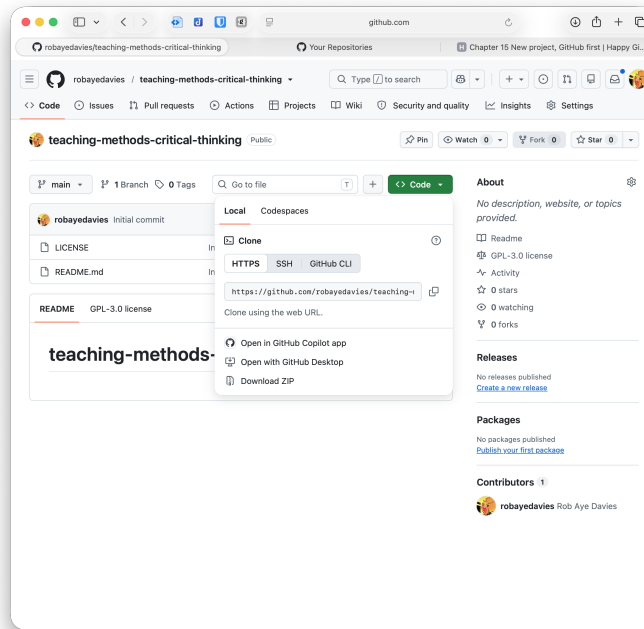


Figure 11.6: github-new-repo-url

In RStudio, enter the URL in the **Clone Git Repository** dialogue box.

Click on **Create Project**.

The project folder (and all its contents) is now on your local hard drive.

11.2.4 Check that our local drive and the remote repository are connected

Now, in RStudio, you can check that your local drive folder and the remote repository on GitHub are connected.

- Open the **README.md**. Edit it. Save the edit.
- Click on the **Git** tab (top right window). Notice that the **README.md** file is now associated with a blue M under Status.

In RStudio > Git

- Click on the button box under **Staged** to get a tick in the box.

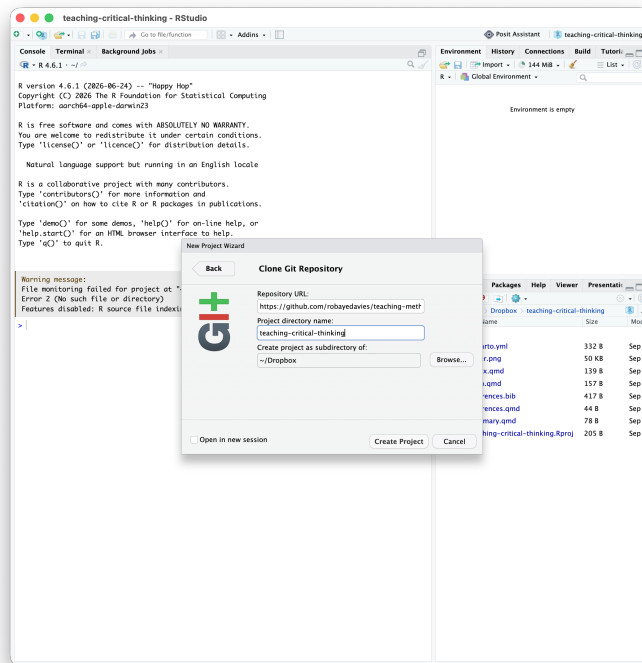


Figure 11.7: rstudio-create-project-version-control-git-url

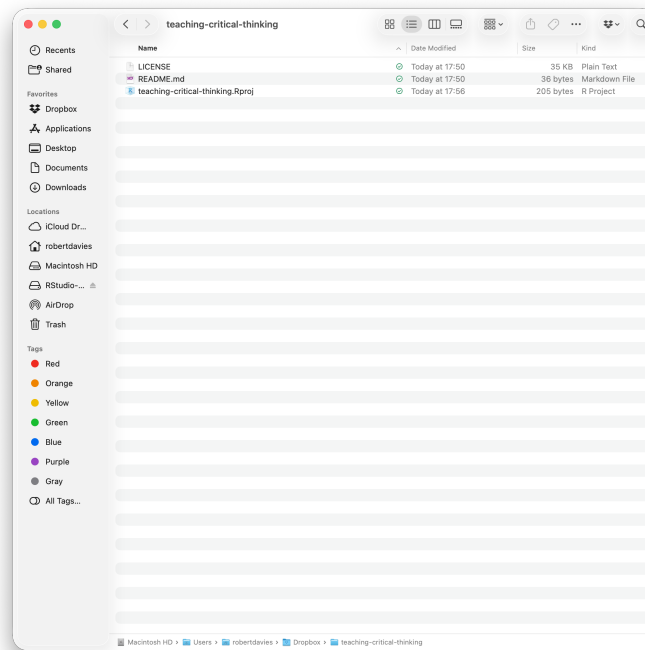


Figure 11.8: mac-finder-project

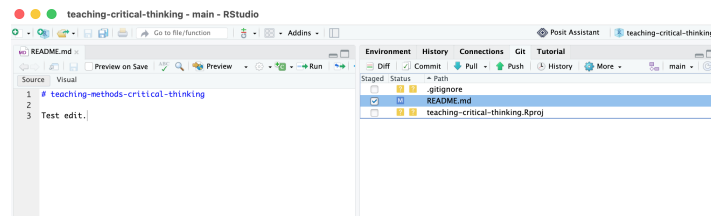


Figure 11.9: rstudio-git-commi

- Click on the **Commit** button above.
- A window opens.
- Review the changes you have staged. Deletions are shown in red. Additions are shown in green.
- Enter a brief but informative message in the commit message box. Click on **Commit**.

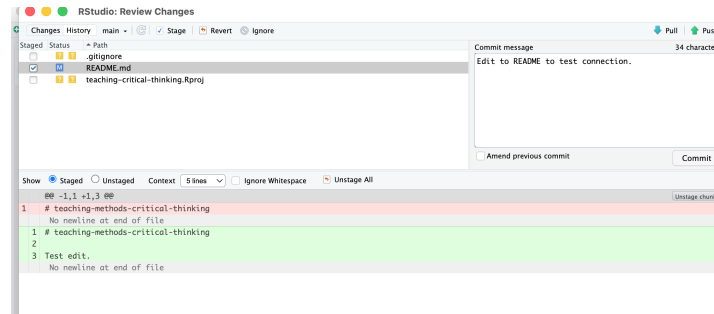


Figure 11.10: rstudio-git-commit-message

- Best practice advice is to commit often, push less frequently.
- RStudio > Git We click on **Push** to test the connection.

Assuming the connection is good, clicking on the **Push** will show you a window showing a summary of what has been changed in (pushed up to) the repo.



Figure 11.11: rstudio-git-commit-message-push

Now, in GitHub, you can check that your local drive folder and the remote repository on GitHub are connected.

- At the repo webpage, you can see the commit message, a with a timestamp.
- This is why it is wise to write informative messages: you want to know what you did when.
- You can click on a file name to see the edits.

That was pushing. How about pulling?

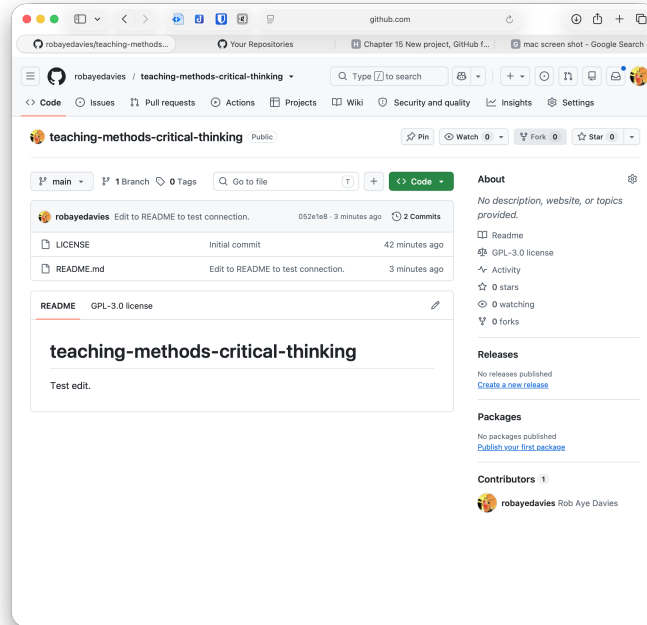


Figure 11.12: github-commits-edits

- Click on the Pull button.

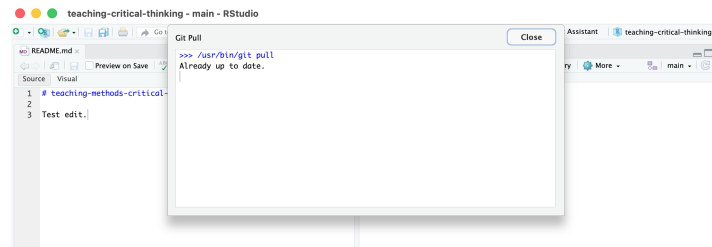


Figure 11.13: rstudio-git-pulln

- It works i.e. you can get changes to the repository downloaded to your local drive but there are no changes and everything is up to date.

11.2.5 Create a new book

Let's say I don't already have a book but I want to create a book, and I want to make a Quarto book. We can create a set of files to act as a skeleton for the book we are going to write.

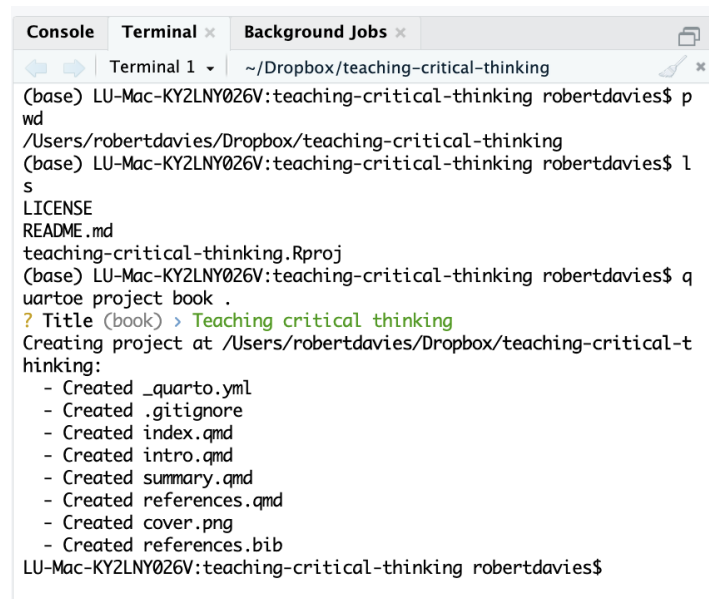
In RStudio, in the terminal, we write the command:

```
quarto create project book .
```

The trailing dot tells Quarto to use the current directory rather than making a new subfolder.

- You will be asked to enter a title.

In the terminal, you will see results that look like this. The directory information will be different.



```
Console Terminal x Background Jobs x
Terminal 1 ~ /Dropbox/teaching-critical-thinking
(base) LU-Mac-KY2LNY026V:teaching-critical-thinking robertdavies$ pwd
/Users/robertdavies/Dropbox/teaching-critical-thinking
(base) LU-Mac-KY2LNY026V:teaching-critical-thinking robertdavies$ ls
LICENSE
README.md
teaching-critical-thinking.Rproj
(base) LU-Mac-KY2LNY026V:teaching-critical-thinking robertdavies$ quarto create project book .
? Title (book) > Teaching critical thinking
Creating project at /Users/robertdavies/Dropbox/teaching-critical-thinking:
- Created _quarto.yml
- Created .gitignore
- Created index.qmd
- Created intro.qmd
- Created summary.qmd
- Created references.qmd
- Created cover.png
- Created references.bib
LU-Mac-KY2LNY026V:teaching-critical-thinking robertdavies$
```

Figure 11.14: rstudio-terminal-quarto-create-book

This bash code (a sequence of code entered at the terminal) creates `_quarto.yml`, `index.qmd`, `intro.qmd`, `summary.qmd`, `references.qmd`, and `references.bib`.

- You can see these files have appeared in the folder you are working with.

11.2.6 Working with the book yaml

In RStudio, in the script window, in the open `_quarto.yml` file, we can see some pre-populated information and the title we entered.

- Edit the yaml to substitute the prepopulated author name with your name.
- Save.

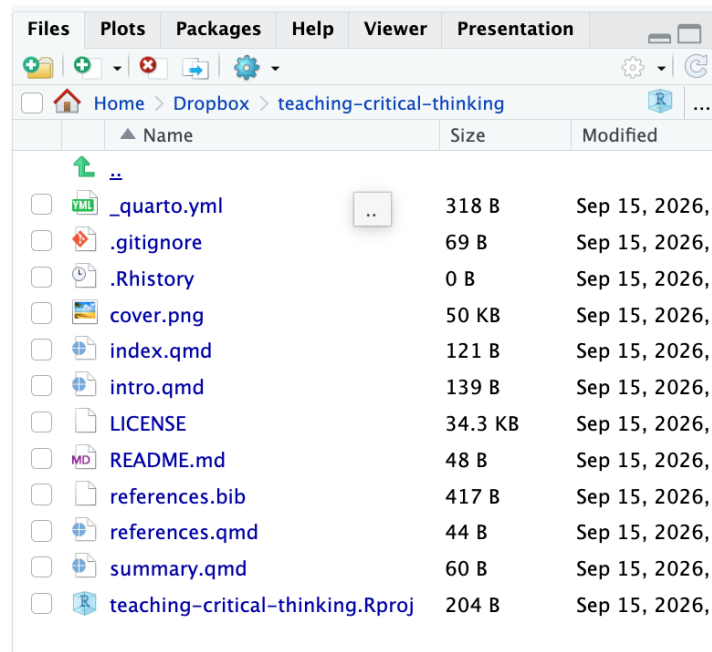


Figure 11.15: rstudio-files-project-book-files

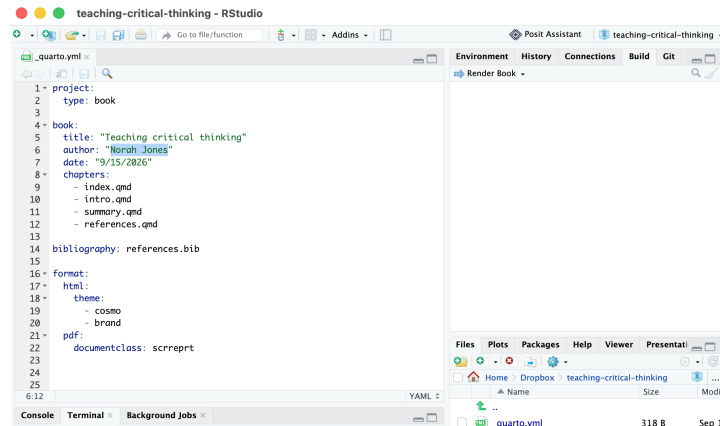


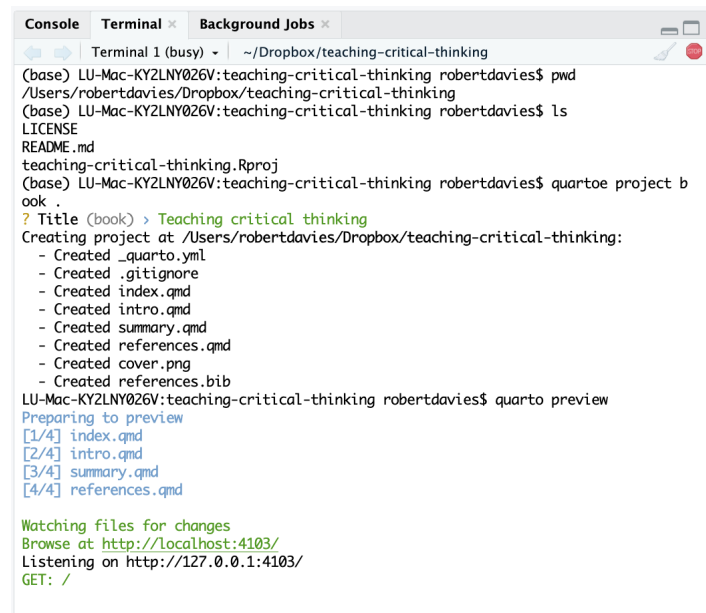
Figure 11.16: rstudio-script-yaml-defaults

11.2.7 Preview what we have got so far.

Click on **Render** in RStudio, or run `quarto preview` in the Terminal.

Your default browser will open a live preview that updates as you edit and save `.qmd` files.

Let's say I run `quarto preview` in the terminal. This is what I see in the terminal.



```
Terminal 1 (busy) ~ /Dropbox/teaching-critical-thinking
(base) LU-Mac-KY2LNY026V:teaching-critical-thinking robertdaves$ pwd
/Users/robertdaves/Dropbox/teaching-critical-thinking
(base) LU-Mac-KY2LNY026V:teaching-critical-thinking robertdaves$ ls
LICENSE
README.md
teaching-critical-thinking.Rproj
(base) LU-Mac-KY2LNY026V:teaching-critical-thinking robertdaves$ quarto project book .
? Title (book) > Teaching critical thinking
Creating project at /Users/robertdaves/Dropbox/teaching-critical-thinking:
- Created _quarto.yml
- Created .gitignore
- Created index.qmd
- Created intro.qmd
- Created summary.qmd
- Created references.qmd
- Created cover.png
- Created references.bib
LU-Mac-KY2LNY026V:teaching-critical-thinking robertdaves$ quarto preview
Preparing to preview
[1/4] index.qmd
[2/4] intro.qmd
[3/4] summary.qmd
[4/4] references.qmd

Watching files for changes
Browse at http://localhost:4103/
Listening on http://127.0.0.1:4103/
GET: /
```

Figure 11.17: rstudio-terminal-quarto-preview

A new webpage tab should open. Here is what Safari shows me:



Figure 11.18: safari-browser-quarto-preview

Notice that in RStudio > Terminal, you can see a red button and the message

```
Watching files for changes
Browse at http://localhost:5105/
```

Listening on `http://127.0.0.1:5105/`
GET: `/`

- Click on the red button and the Preview stops.

This preview is local to you. In the next step, we publish what we have got so far so anyone can view it, if they have the URL.

There are two ways to do this:

1. The simplest: run `quarto publish gh-pages` from the Terminal. This renders the book, pushes it to an auto-created `gh-pages` branch, and **GitHub Pages** picks it up automatically.
2. Alternative: set `'output-dir: docs'` in `__quarto.yml`, render, commit the `docs/` folder, add a `.nojekyll` file, and point **GitHub Pages** at the `docs/` folder in **Settings > Pages**.

I am going to use the second route.

11.2.8 Publish your new book to live webpages

In Rstudio, in the script window, in `__quarto.yml`, set the output directory.

- Edit the `yml`, adding on a new line: `output-dir: docs`.

In Rstudio, in the terminal, enter `touch .nojekyll`.

Add a `.nojekyll` file to your repo root (`touch .nojekyll` in Terminal, or copy `NUL .nojekyll` on Windows). This stops **GitHub** from running its own **Jekyll** processor over your site, which can break **Quarto's** output.

In Rstudio, in the terminal, `quarto render`.

- The terminal will display its progress through the rendering process.
- A review of the contents of the folder will show the new `/docs/` folder.

In Rstudio, in git, commit all the new files. Then push.

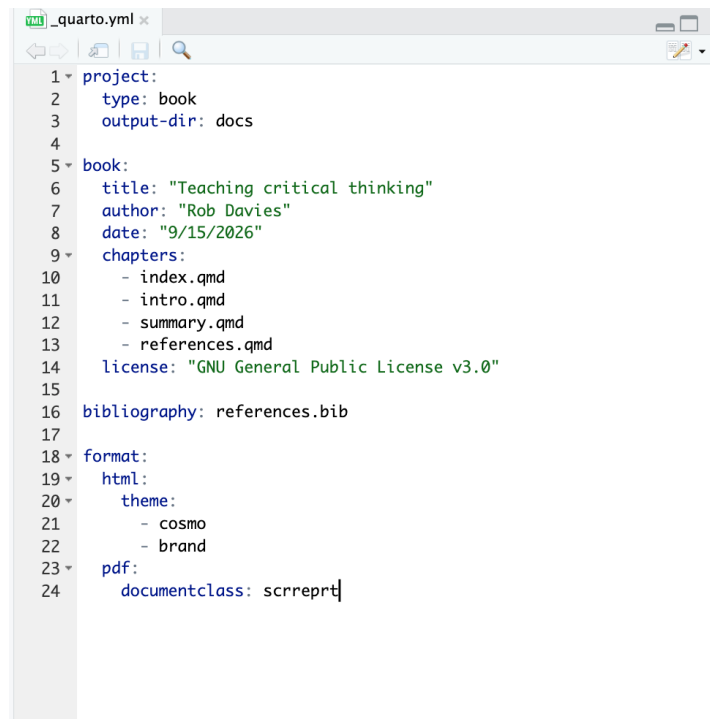
Commit all new book files.

11.2.8.1 Fix the license

We might not have already done this but we need to choose a license for our materials.

In **GitHub**, change the license to **CC.0**

Before doing anything else, I pull this remote change to the local version.



```
1 project:
2   type: book
3   output-dir: docs
4
5 book:
6   title: "Teaching critical thinking"
7   author: "Rob Davies"
8   date: "9/15/2026"
9   chapters:
10    - index.qmd
11    - intro.qmd
12    - summary.qmd
13    - references.qmd
14   license: "GNU General Public License v3.0"
15
16 bibliography: references.bib
17
18 format:
19   html:
20     theme:
21       - cosmo
22       - brand
23   pdf:
24     documentclass: scrreprt
```

Figure 11.19: rstudio-script-yaml-book-docs

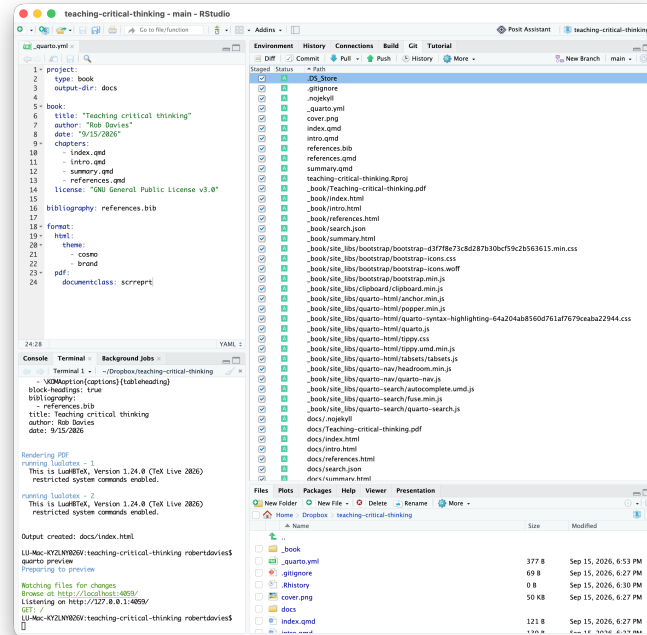


Figure 11.20: rstudio-git-commit-new-all

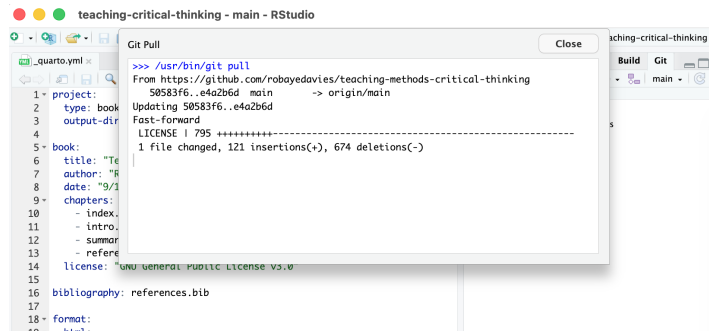


Figure 11.21: rstudio-git-pull-license-edit

11.2.8.2 Publish the book

Point GitHub Pages at `/docs`

On GitHub.com, go to your repo's Settings > Pages. Under Build and deployment, set Source to Deploy from a branch, Branch to main, and folder to `/docs`. Save.

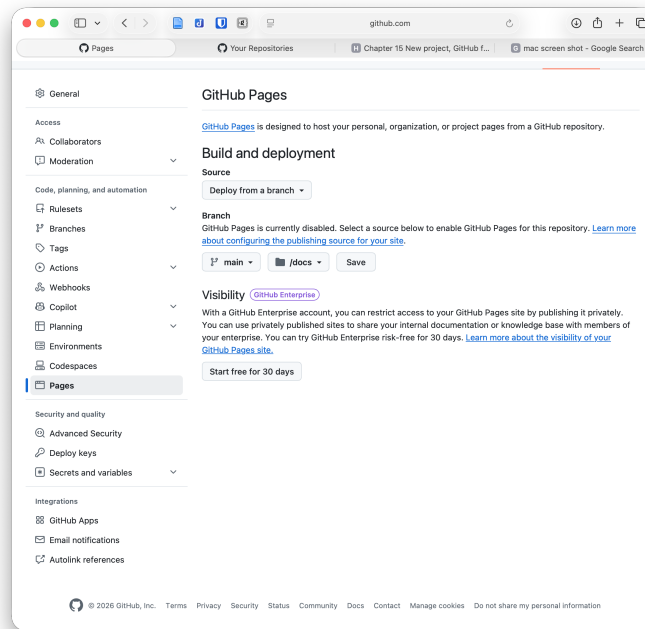


Figure 11.22: github-pages-point-to-docs

11.2.8.3 Check the live site

GitHub will build and deploy within a minute or two. Your site will be live at <https://github.io/>. Any time you edit content, re-run `quarto render`, commit the updated docs/ folder, and push — no separate publish command needed with this method.

- You can see that the book has now deployed to a live webpage in the Deployments column of the GitHub view of your repository.
- If you click on the button `github-pages`, you get taken to a deployment information window:
- where you can see a link (shown as the URL) to the now live webpage:

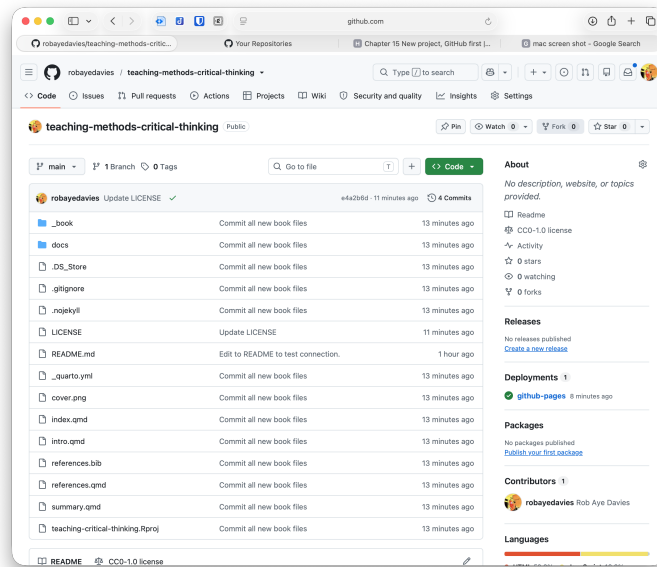


Figure 11.23: github-deployment

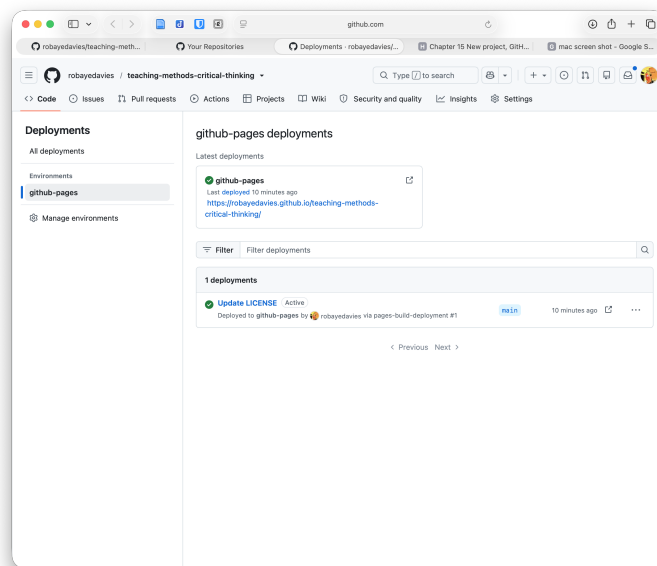


Figure 11.24: github-deployment-info

<https://robayedavies.github.io/teaching-methods-critical-thinking/>

Let's move next to write or to make edits to some content.

12 Quarto workflow overview

13 Introduction working with Quarto

Part III

References

References

Knuth, Donald E. 1984. “Literate Programming.” *Comput. J.* (USA) 27 (2): 97–111.
<https://doi.org/10.1093/comjnl/27.2.97>.